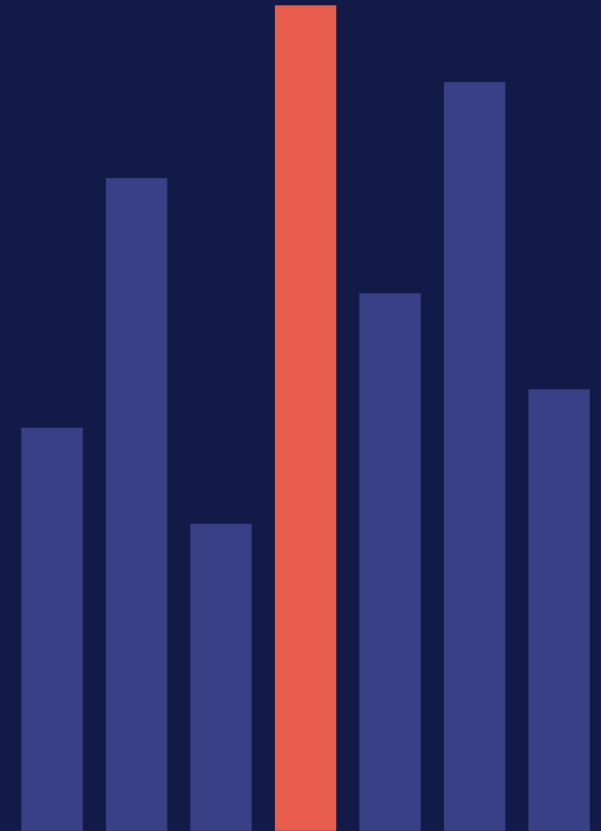


EXPLORATORY DATA ANALYSIS

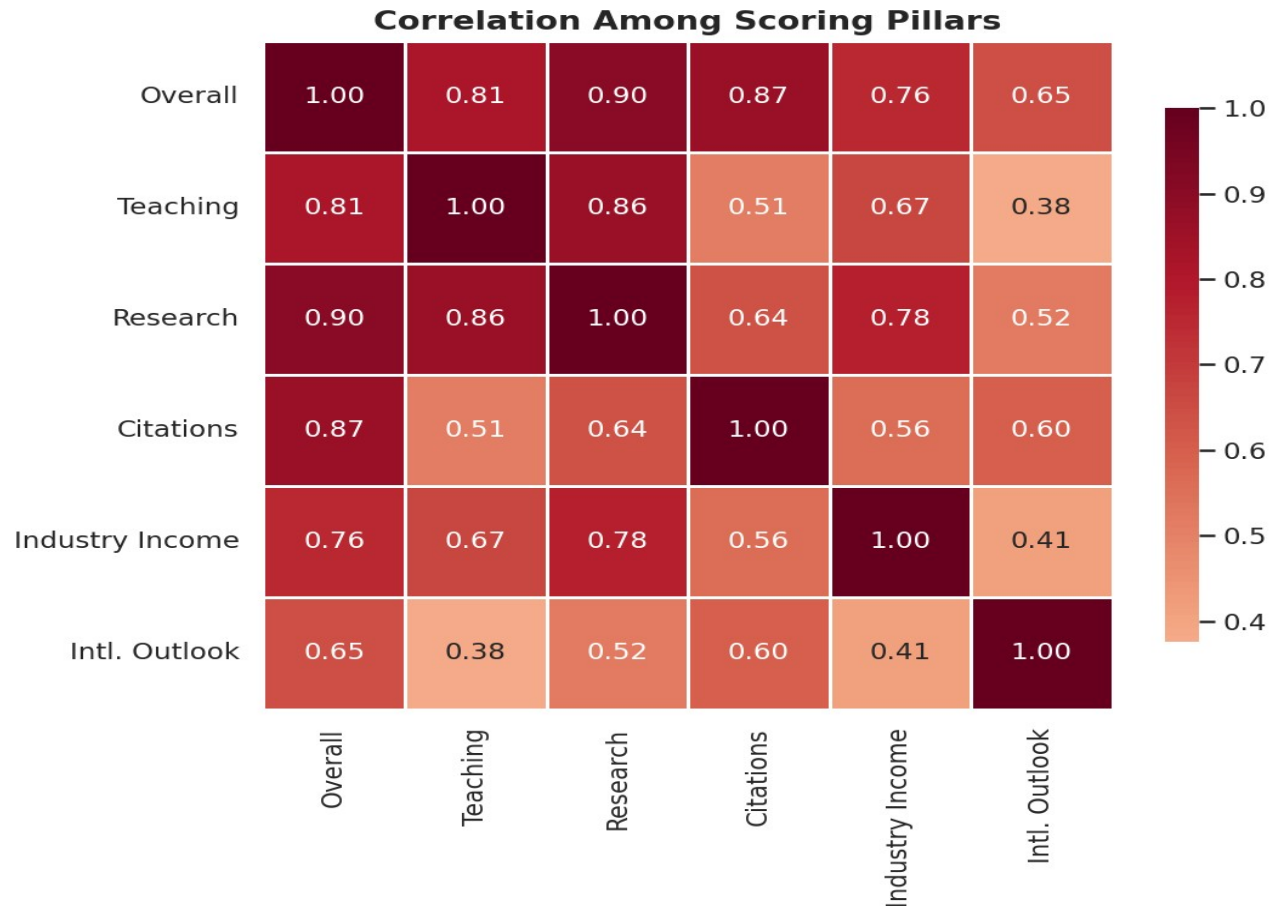
World University Rankings

What actually separates elite institutions from everyone else

Times Higher Education World University Rankings · 2,191 institutions · 115 countries



Research is the dominant driver of overall score



CORRELATION WITH OVERALL SCORE

Research **0.90**

Single strongest predictor

Citations **0.86**

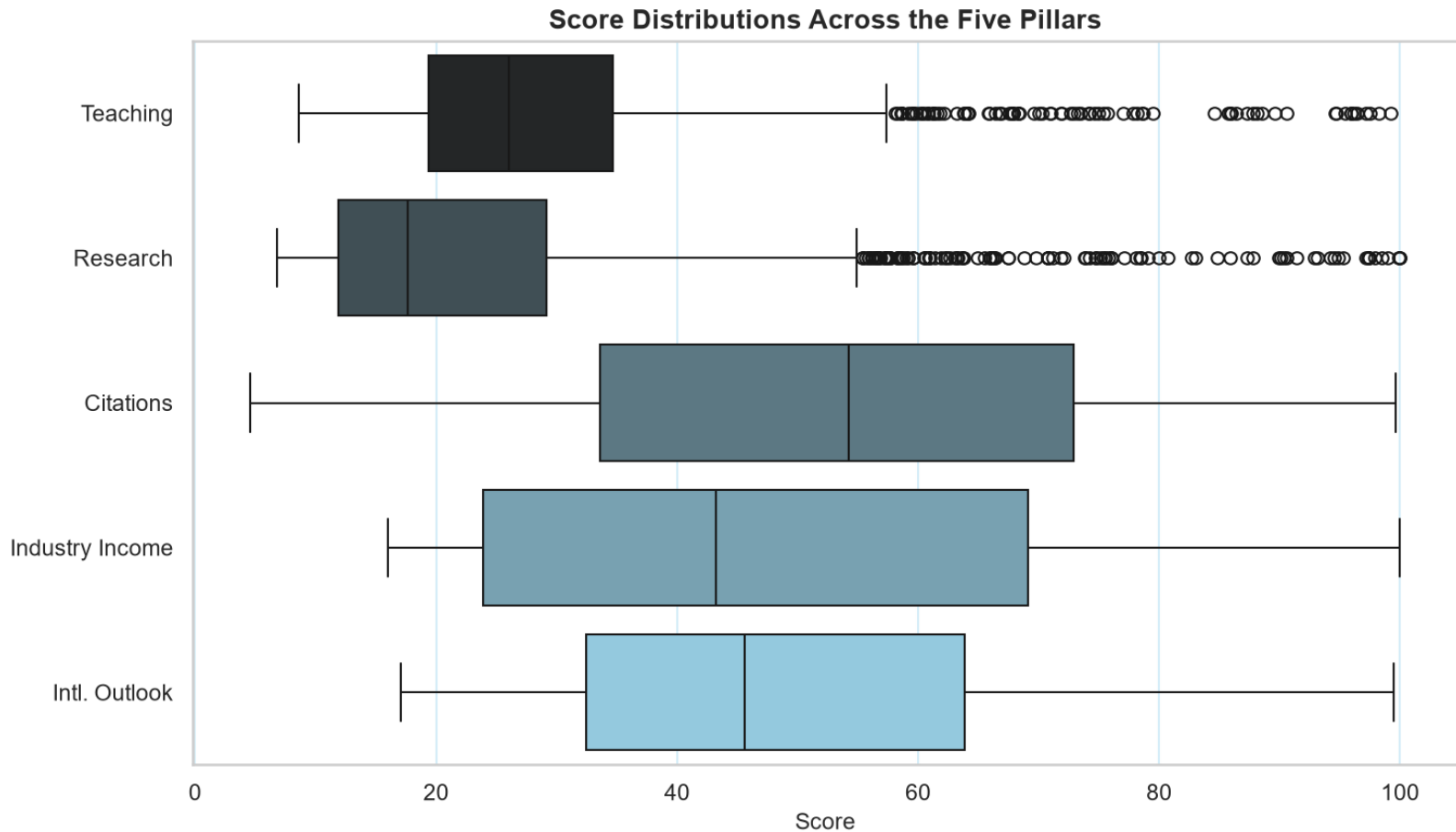
Close second

Teaching **0.81**

Still tightly linked to score

These are correlations among the five scoring pillars and the overall score only — institutional traits like staff ratio and international mix are covered separately.

The five pillars are not on a comparable footing



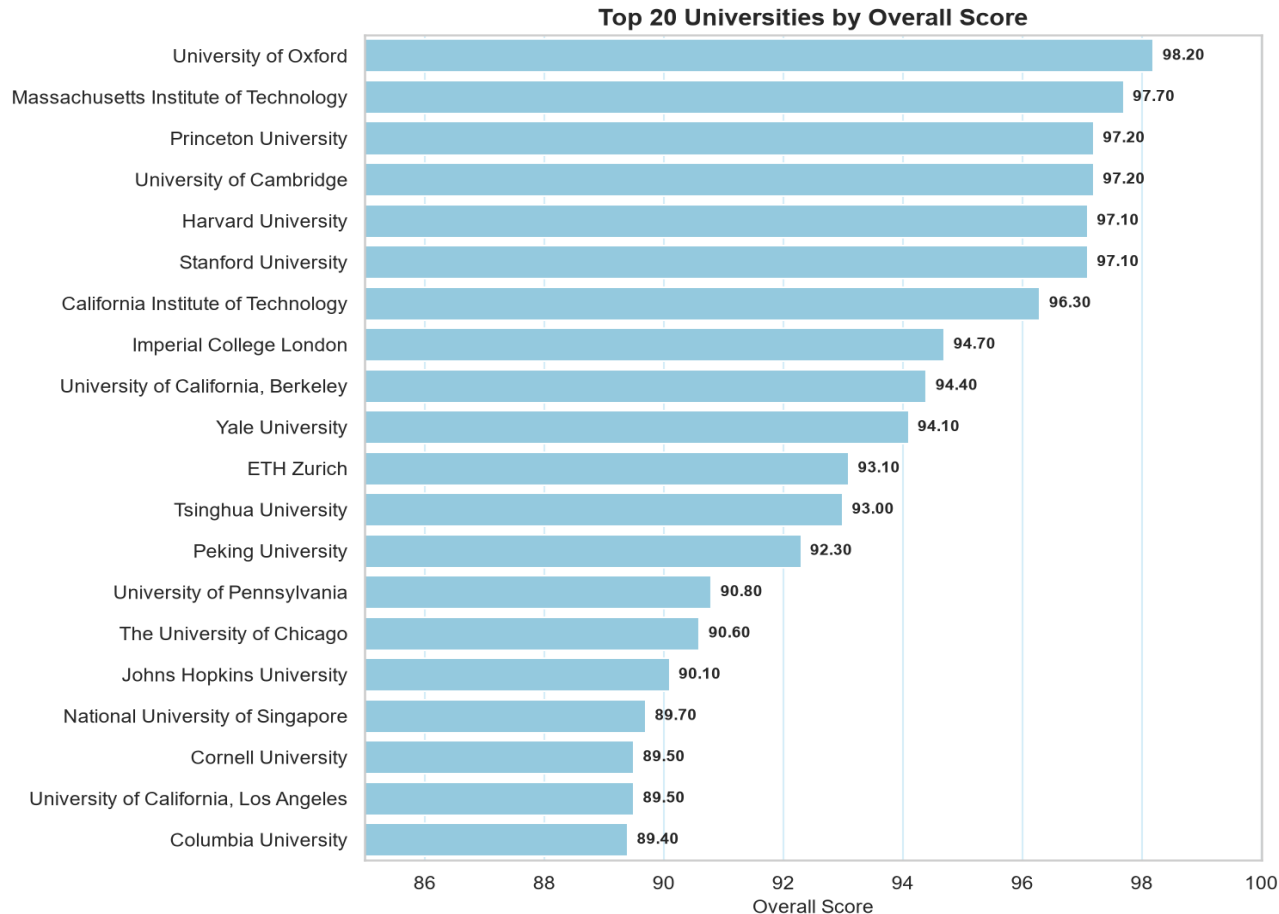
Citations run hot

Citations score far higher on average than Research or Teaching — a mid-tier school can post a strong citations score while being nowhere near research-competitive.

Compressed vs. spread out

Citations & Industry Income spread broadly and separate institutions poorly. Research & Teaching sit compressed at the bottom with long elite tails.

A very tight cluster at the very top



A 1.9-point spread

Oxford (#1, 98.2) to Caltech (#7, 96.3) — the leading pack is a tight cluster fighting over fractions of a point, not a steep drop from a lone leader.

WHO MAKES THE TOP 20

13

United States

of 171 ranked US universities

3

United Kingdom

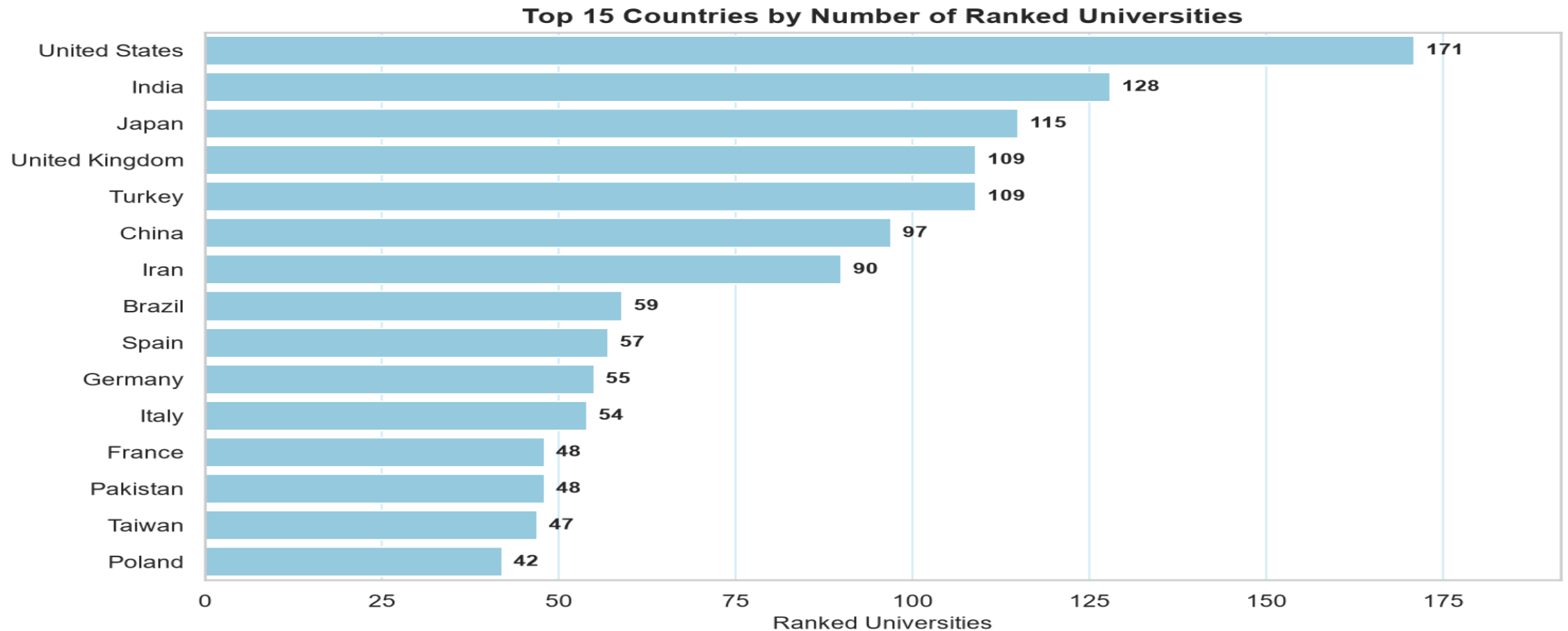
of 109 ranked UK universities — Oxford, Cambridge, Imperial

4

Rest of world

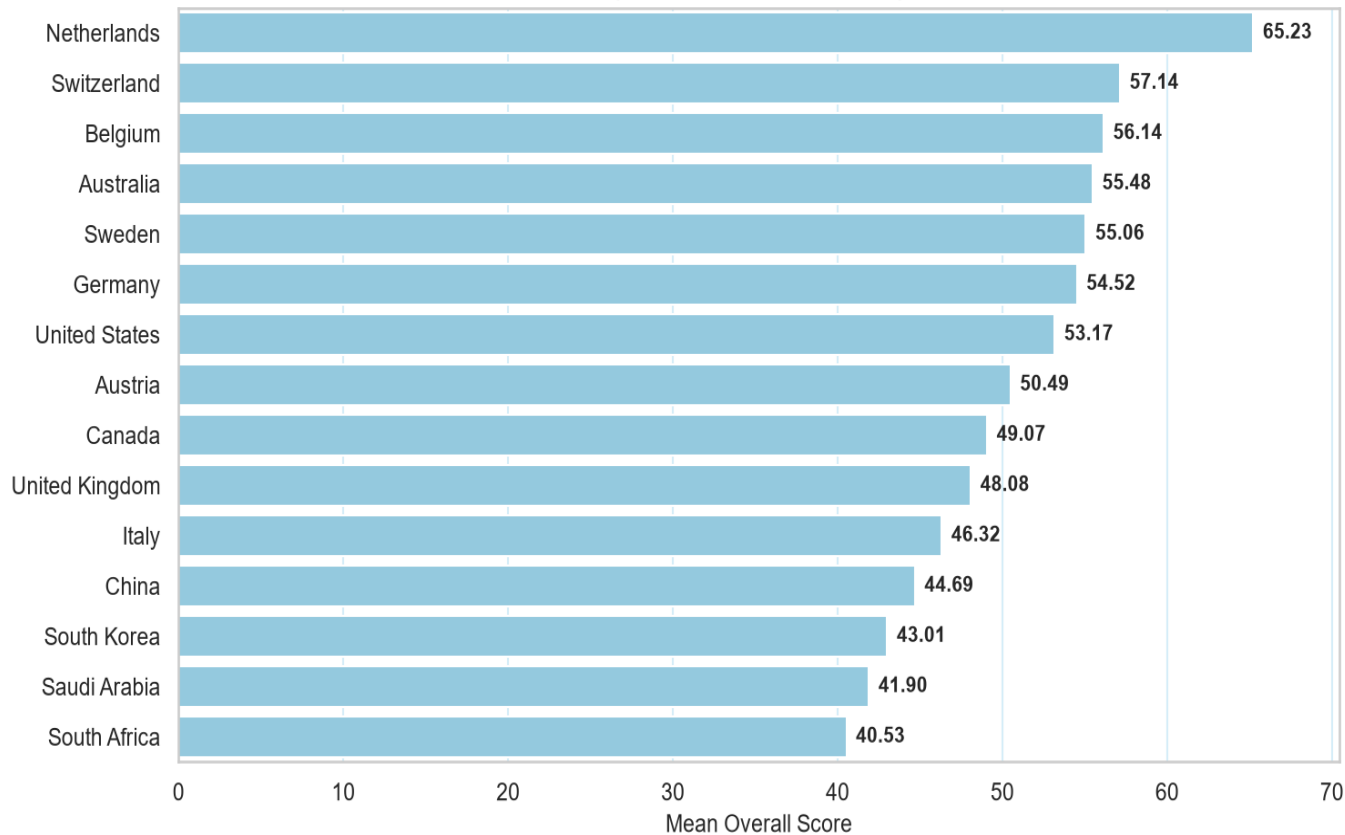
ETH Zurich, Tsinghua, Peking, NUS

Who has the most ranked universities?



Volume and quality decouple sharply

Top 15 Countries by Average Overall Score
(10+ ranked universities)



Min. 10 ranked universities per country, so one or two outliers can't top the table.

Netherlands

65.2

#1 quality

Switzerland

57.1

Belgium

56.1

United States

53.2

171 unis, ranks 7th

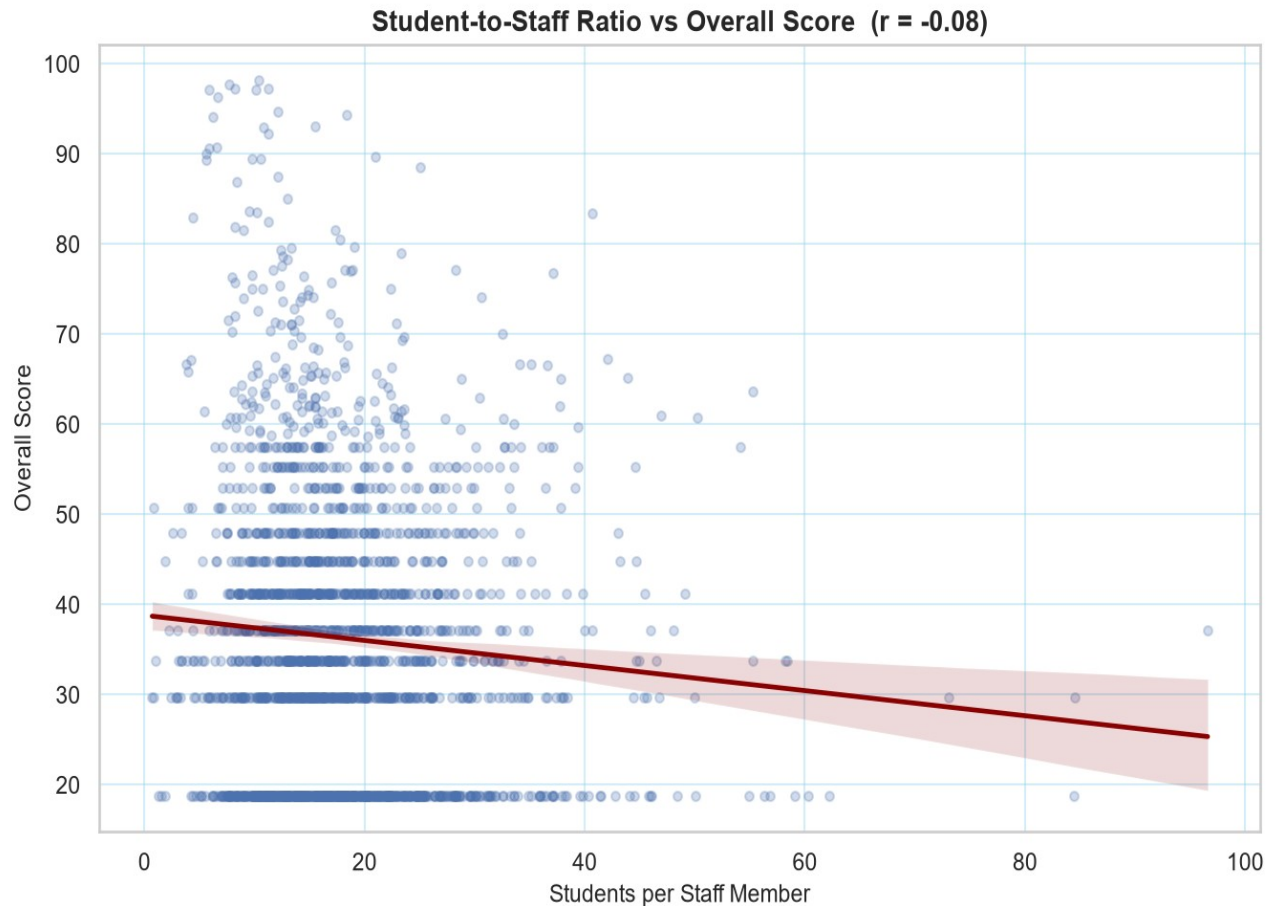
United Kingdom

48.1

109 unis, ranks 10th

Large systems rank many mid-tier schools alongside their flagships, pulling the average down — headcount alone is a poor quality proxy.

Student-staff ratio is not a useful quality signal



$$r = -0.076$$

ESSENTIALLY UNCORRELATED WITH OVERALL SCORE

Small classes are a conventional proxy for teaching quality. The data doesn't support that assumption at the institutional level — though the trendline does show a slight dip in score as students-per-staff increases.

KEY TAKEAWAYS

Conclusions

1

Research is the dominant driver of rank

$r = 0.90$, and the largest top-100-vs-rest gap (+51.9 points). Elite status is fundamentally a research phenomenon.

2

Internationalization may be a promising lever

$r = 0.52$ (share of international students) is moderate, so it may prove a good aspect to build: diversity can be grown through admissions and recruitment far faster than research capacity.

3

Breadth and quality decouple nationally

The US (171 unis) and UK (109) lead on count but sit 7th (53.2) and 10th (48.1) on average score, behind the Netherlands (65.2), Switzerland (57.1) and Belgium (56.1) — restricted to countries with 10+ ranked universities.

4

Pillars aren't equally discriminating

Citations and Industry Income are broadly distributed and separate institutions poorly; Research and Teaching are compressed at the bottom with long elite tails.

5

Student-staff ratio is not a useful quality proxy

$r = -0.076$, essentially uncorrelated with overall score — despite being a common admissions talking point.